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# Student Project Hazard Assessment

A Hazard Assessment is designed to help students and project advisors recognize hazards associated with student projects at the early planning stages to find ways to minimize the chance of injury, loss, or harm while students are working on the project. A detailed assessment will also prevent the potential for project interruption due to unanticipated safety issues. This form is intended for projects where the primary hazards are associated with engineering work (physical, mechanical, electrical, etc.). Chemical and biological focused projects require a separate project assessment form.

Each student project must complete a Hazard Assessment form, *even if the project is not expected to have any hazards*. All required approvals must be obtained before proceeding with the project. It is important that all team members participate in the process, with close supervision of your advisor. To help ensure that hazards and risks associated with your project are not overlooked or underestimated, you are encouraged to contact additional university staff (lab directors/managers, EHS, etc.) with relevant knowledge or experience for guidance.

The Hazard Assessment process usually involves these five steps below, with an example:

| Step: | Example: |
|---|---|
| 1. **Identify the specific tasks** that must be completed to reach your project goals | One of your project tasks involves testing a live electrical circuit |
| 2. **Determine if there are hazards** associated with completing the tasks | On the form, you select the "Electrical parts and assemblies > 50V or high current", under the Hazardous Conditions/Processes/Activities, Electrical Hazard section |
| 3. If hazards exist, **identify the risks** connected with the hazards of each task. Ask yourself, what could go wrong? If you are not already familiar with the risks, do a quick internet search | After some research, you learn that there is potential of electrical shock from accidental contact with exposed live components |
| 4. **Develop a list of controls** (things you can do) to eliminate the hazard or reduce the risks. Refer to *Hierarchy of Controls* on the next page | To minimize the risk identified above, you could:<br>o De-energize and isolate the system or<br>o Guard live components to prevent accidental contact |
| 5. **Create a safe working procedure**. Describe how you will safely complete each task | You write a detailed procedure for testing a live electrical wire, that includes all the information from your hazard/ risk assessment and which controls you will use to reduce the risks |
| 6. **Submit your hazard assessment form for approval (see section below for details)** | |

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## Definitions

A hazard is something that has the potential to cause harm (injuries, accidents, or other undesirable effects). Hazards can be eliminated but not reduced. A hazard can be in the form of an agent, condition, process or activity.

Risk is the likelihood or probability of a hazard causing harm to people, property, or the environment. Risks associated with a hazard can be reduced. Put another way,

*Risk = Hazard x Exposure*

A hazard assessment is the process of identifying anything that can cause harm (hazardous agents, conditions, processes or activities). A risk assessment is the process of determining how great the chance is of harm occurring from a given hazard.

## Hierarchy of Controls

Unless the hazardous agent, condition, or activity is removed, hazards cannot be eliminated. However, risks from the hazard can be minimized by employing the proper control measures and safe work practices that will have been identified from completing a hazard assessment.

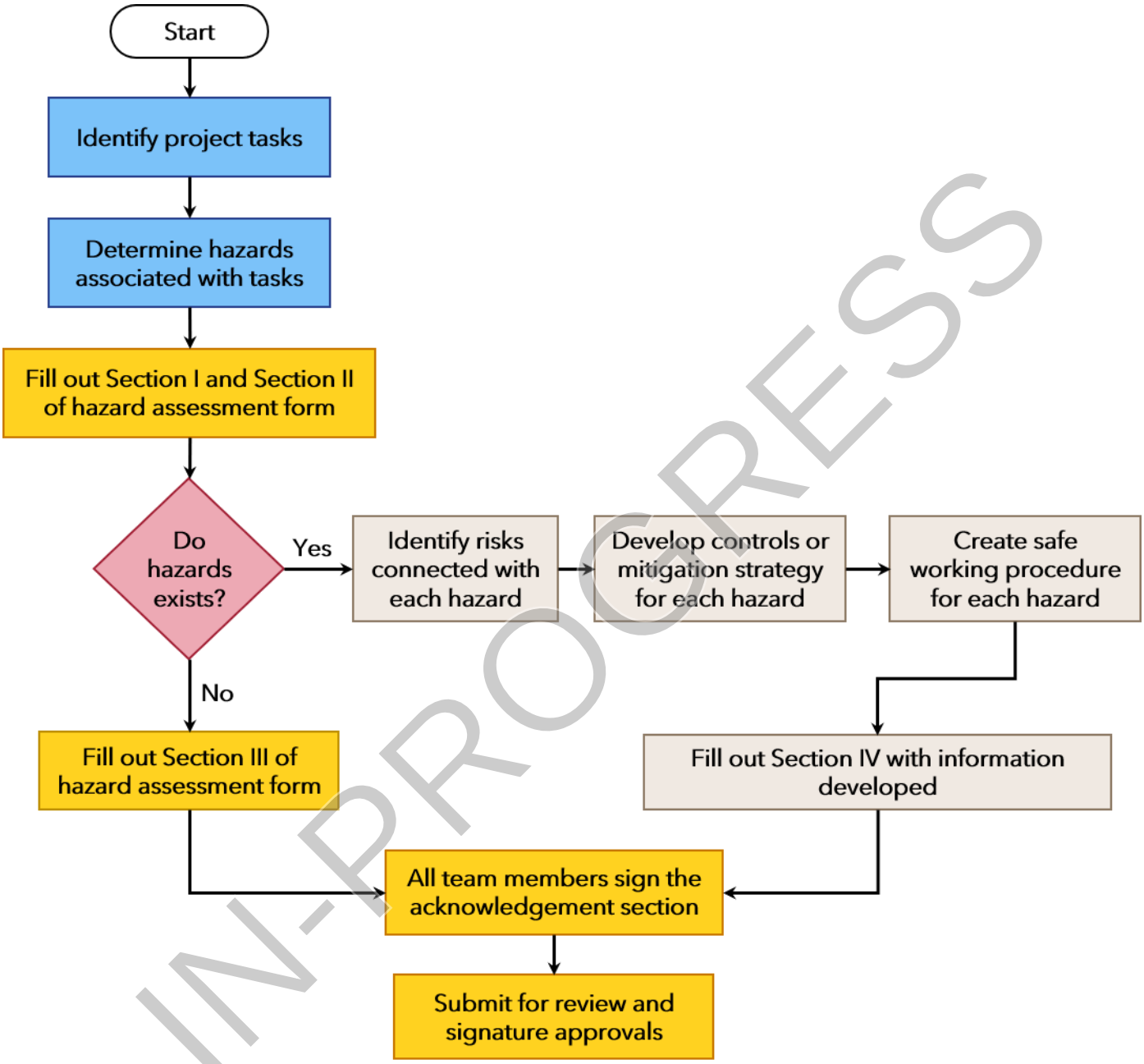
Some controls are more effective than others at eliminating hazards or reducing risk. Use the hierarchy of controls chart below to evaluate the controls measures you plan to use. Priority should be given to the most effective controls at the top of the hierarchy (elimination and substitution) and moving down, rather than start with the easiest one. While personal protective equipment (PPE) should always be used, it should be considered the last line of defense from potential hazards.




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## Hazard Assessment Form Process




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1. After completing the form, all team members must sign the acknowledgement section.
2. Your advisor and the department chair must approve this form prior to obtaining formal approvals from other university staff.
3. If your project involves the use of lab or shop space or equipment, the form must be routed to the appropriate lab/shop manager(s) for review and approval.
4. Your faculty advisor or department chair will make the determination whether EHS approval is additionally required.

Depending on the scope of your project, the faculty advisor, department chair, lab/shop manager(s), or EHS may request further approval from other relevant faculty, staff, or department.

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# Student Project Hazard Assessment Form

This form is to be used for student projects where primary hazards are associated with engineering work (physical, mechanical, electrical, etc.). Chemical and biological focused projects require a separate form.

Complete this form and obtain all the required approvals (Faculty Advisor, Department Chair, Laboratory Manager, EHS, etc.) before proceeding with the project. Please refer to the hazard assessment guide for assistance in filling this form.

## SECTION I: Project Description

Project Title:
S.C.O.U.T. (Santa Clara Oceanic Utilities Transmitter): Low-Cost Modular Nearshore Environmental Monitoring Buoy

Project Team Members:
John Ryan Myrdal (GENG) · David Chousal Cantu (CSEN) · Isabella Rodriguez (ECEN)

Project Faculty Advisor

| Name: | Department: | Phone: | Email: |
|---|---|---|---|
| Jes Kuczenski | GENG | | jkuczenski@scu.edu |

Proposed Project Location(s) (Department, building, room#):
EPIC Lab, Bergin 205

Anticipated Dates of Project Duration:
Aug 14, 2026 to May 28, 2027. On-campus build and test runs about Oct 2026 to Feb 2027.

Summary of Project Objectives:
S.C.O.U.T. is a low-cost, battery-powered, modular buoy. It measures water temperature and turbidity, and may later record hydrophone audio, to monitor coral-reef health. The team builds a 3D-printed buoy, sealed electronics and a LoRa link to a shore station, then tests it in lab bins, a pool or pier, and open water. The final deployment is in Kaneohe Bay, Hawaii (Mar to May 2027), subject to permits.

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# SECTION II: Hazard Checklist

Identify all the tasks that must be completed for your project. Carefully evaluate each task to determine if there are any associated hazards. After identifying the hazards of your project, you will be asked to assess the risk connected to each hazard and to identify control measures that will either eliminate the hazard or reduce the risk to an acceptable level. Safe work procedures for each step involving a known hazard will need to be developed.

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## HAZARDOUS CONDITIONS / PROCESSES / ACTIVITIES

### Electrical Hazards

☐ Electrical parts and assemblies > 50V or high current

☒ Batteries

☐ Control Panels

### Mechanical Hazards

☒ Power tools and equipment

☒ Machine guarding/power transmission: gears, rotors, wheels, shafts, belt/chain drives, rotating parts, pinch points

☐ Robotics

☒ Sharp Objects

☒ Stored Energy (springs, gravity, pneumatic, hydraulic, pressure)

### Physical Hazards

☒ Extreme temperature (high temp fluids: water > 160 °F, steam, hot surfaces > 140 °F, cryogenic fluids

☒ Material handling of heavy objects

☐ Elevated heights (scaffolding, ladders, roofs, lifts, etc.)

☐ Overhead falling objects (cranes, hoists, drones, projectiles, etc.)

☐ Confined Spaces

☒ Airborne Dusts

☐ Bonding / Grounding

☒ Electrostatic Discharge

### Reaction Hazards

☐ Explosive

☐ Exothermic, with potential for fire, excessive heat, or runaway reaction

☐ Endothermic, with potential for freezing solvents, decreased solubility or heterogeneous mixtures

☐ Gases produced

☐ Hazardous reaction intermediates/products

☐ Hazardous side reactions

### Hazardous Processes

☒ Generation of air contaminants (gases, aerosols, or particulates)

☐ Heating chemicals

☒ Large mass or volume

☐ Pressure > Atmospheric

☐ Pressure < Atmospheric

☐ Scale-up of reaction

☒ Metal fabrication (soldering, welding, cutting, drilling, etc.)

☒ Construction/Assembly, etc.

### Other Hazards

☒ Noise > 80 dBA

☐ Vehicle traffic

☐ Hazardous waste generation

☒ Other (list):

Water work from a pier, boat slip or open water (drowning, falls)

☐ Check here if project will not involve any hazardous conditions, processes, or activities

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Hazard Checklist (continued)

| HAZARDOUS AGENTS | | | |
|---|---|---|---|
| **Physical Hazards of Chemicals** | **Health Hazards of Chemicals** | **Non-Ionizing Radiation** | **Biohazards** |
| ☐ Compressed Gases | ☒ Acute Toxicity | ☐ Lasers | ☐ BSL-2 Biological Agents |
| ☐ Cryogens | ☐ Carcinogens | ☐ Magnetic Fields (e.g. NMR) | ☐ rDNA |
| ☐ Explosives | ☐ Nanomaterials | ☒ RF/Microwaves | ☐ Human Cells, Blood, BBP |
| ☒ Flammables | ☐ Reproductive Toxins | ☐ UV Lamps | ☐ Animal Work |
| ☐ Oxidizers | ☒ Respiratory or Skin Sensitization | | |
| ☐ Peroxides or Peroxides Formers | ☐ Simple Asphyxiant | | |
| ☐ Pyrophorics | ☒ Skin Corrosion/ Irritation | | ☐ Other (List): |
| ☒ Water Reactives | ☐ Hazards Not Otherwise Classified | | |
| ☐ Check here if project will not involve any hazardous agents | | | |

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# SECTION III: Projects with No Hazards

IF your project meets all these conditions:

- ☐ No hazardous conditions, processes, or activities
- ☐ No hazardous agents
- ☐ Will not use any laboratory or shop space

You can stop at this section. Acknowledge below and obtain the approval from your faculty advisor.

**If your project involves hazardous processes, activities, conditions or agents are, please continue to SECTION IV to complete the rest of the form.**

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## ACKNOWLEDGEMENT (for projects with no hazards)

By signing, I verify that the project will not involve any hazards listed in Section II

| Name of Project Team Member | Signature | Date |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |

## APPROVALS (for projects with no hazards)

This document must be reviewed and approved by the people below before any project work can begin. A copy of the approved document must be kept with the project team

**Faculty Advisor** (required)

Name:

| Department: | Signature | Date |
|---|---|---|
| | | |

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# SECTION IV: Hazard Assessment and Control

## Description of Potential Hazards

Provide a summary of the procedure and describe the risks associated with each hazard that you have identified above or on the previous page. Use one box below per hazard. You may add additional pages if needed. Define the hazard control measures that will be employed to minimize the risks based on the hierarchy of controls (elimination, substitution, engineering controls, administrative controls, PPE), and then describe specific control measures you will use (e.g. work on system de-energized, receive hazard specific training, shield hot surfaces, guard pinch points, relieve stored energy, wear protective equipment, use less hazardous chemical, etc.). Refer to "Hierarchy of Controls" in the instructions sheet for more information to decide which hazard controls measures are most appropriate.

| **Hazardous Activity, Process, Condition, or Agent** (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Batteries (two Saft LS33600 Li-SOCl D cells). Bench-test on a power supply first. Install the cells in the sealed battery bay. Replace at end of life. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>A shorted, crushed, punctured or charged cell can leak or vent toxic thionyl chloride, overheat or ignite. Lithium reacts with water if breached. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Use a bench supply during development. No charge path to the cell; USB link verified isolated. Insulated tools. Read the Saft datasheet and SDS. Safety glasses and nitrile gloves. Fire extinguisher nearby. Spent or damaged cells are never put in trash or recycling: tape the terminals, store in a labeled non-metal container, and hand to Environment, Health and Safety (EHS) for hazardous waste pickup. |

| **Hazardous Activity, Process, Condition, or Agent** (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Solder boards, headers and sensor cables at a ventilated station. Handle boards on an ESD mat. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Burns from the iron. Solder flux fumes. Electrostatic discharge (ESD) can damage boards and sensors. Fire from an unattended iron. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Done in the EPIC Lab, a controlled lab, [under lab manager oversight]. Fume extractor at the station. Lead-free solder. Iron holder and heat mat. Unplug when unattended. Safety glasses. ESD mat and wrist strap. Electrical Safety for Engineering course. |

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| Hazardous Activity, Process, Condition, or Agent (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Print buoy parts in PETG on lab 3D printers. Press in heat-set inserts. Assemble the buoy. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Burns from the hot end, bed and insert iron. Gantry pinch points. Heated PETG releases fumes (volatile compounds and fine particles) while printing. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Print only in [approved lab / approved open space with ventilation]. Printer enclosed or vented where possible. Do not stay in the room for long prints. Let the hot end and bed cool before touching. Keep hands out of the gantry while it moves. Insert iron on a holder, not unattended. Safety glasses. |

| Hazardous Activity, Process, Condition, or Agent (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Drill, trim and sand printed parts. Remove supports with blades. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Our own process generates the dust: drilling, trimming and sanding printed PETG throws off airborne plastic dust and chips. Inhalation and eye irritation. Cuts from blades and support removal. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Done only in [approved lab or approved open space]. Clamp the work. Cut away from the body. Work where the dust is captured or ventilated. Safety glasses, cut-resistant gloves and a dust mask. Wet-wipe or HEPA-vacuum dust, not dry sweeping or compressed air. Lab-specific tool training. |

| Hazardous Activity, Process, Condition, or Agent (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Run float, leak and soak tests in bins and tanks with synthetic seawater. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Shock or short from water around powered electronics. Slips on wet floors. Salt damage to equipment. Bin leaks or overflow: a leaking bin can spill 50 gal or more of synthetic seawater, and rooms without floor drains let it spread to other equipment and rooms. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Bins sit [in a room with a floor drain / on spill containment such as a pallet or tub; confirm which location]. No bins in rooms without a drain unless on containment. Do not overfill. Check bins and seams for leaks at setup and daily. Keep mains power away from tubs and use a GFCI outlet. Run powered electronics only when sealed. Wipe spills at once. Rinse equipment. Keep spill supplies close. |

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| Hazardous Activity, Process, Condition, or Agent (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Move water-filled bins (up to a 50-gal bin, about 400 lb), the buoy and ballast. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Back injury, dropped loads, spills. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Drain bins by pump or siphon before moving. Two-person lift over about 40 lb. Use carts. Keep paths clear. |

| Hazardous Activity, Process, Condition, or Agent (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Seal printed seams and fittings with epoxy and J-B Weld. Clean surfaces with isopropyl alcohol. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Epoxy and isopropyl alcohol give off volatile compounds (VOCs): the alcohol evaporates, and epoxy releases fumes while mixing and curing. Vapors build up in closed spaces, can irritate eyes, skin and lungs, and the alcohol vapor is flammable. Chemical waste. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Work in a fume hood or a well-ventilated area, [confirm which]. Never in a closed or confined space. Keep solvents away from ignition sources. Nitrile gloves and safety glasses. Read the SDS. Collect waste for Environment, Health and Safety (EHS) pickup. |

| Hazardous Activity, Process, Condition, or Agent (identified from previous page): |
|---|
| Summary of Procedure or Tasks:<br>Attended tests at a pier, boat slip or open water, then the Hawaii deployment. A 915 MHz LoRa transmitter is on board. Buoy launch and recovery from the shore or a boat. |
| Describe Hazards (why is the procedure hazardous or what can go wrong – what is the risk):<br>Drowning or falls from the pier or boat. Weather and sea state. Cold or heat exposure. Radio-frequency (RF) exposure from the 915 MHz LoRa transmitter; the transmitter can be damaged if run with no antenna. |
| Hazard Control Measures (what you will do to eliminate the hazard or minimize risks):<br>Never test alone. Wear a PFD. Check weather. Get host or permit approval first. Never transmit without an antenna attached. |

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## SAFETY EQUIPMENT and PPE

**Select the appropriate PPE and safety supplies you will need for the project (Check all that apply)**

☒ Appropriate street clothing (long pants, closed-toed shoes)

☒ Gloves; indicate type: Nitrile

☒ Safety glasses/ goggles

☐ Face shield and goggles

☐ Lab coat

☒ Hearing protection

☒ Fire extinguisher

☐ Eyewash/safety shower

☐ Spill kit

☒ Other (list): PFD/Life Jacket, dust mask (N95)

## TRAINING REQUIREMENTS

**Identify the appropriate training (check all that apply)**

☐ Biology & Bioengineering Lab Safety Camino Course – contact Lab Manager or EHS to enroll

☐ Chemistry & Biochemistry Lab Safety Camino Course – contact Lab Manager or EHS to enroll

☒ Dry Lab Safety Camino Course – contact Lab Manager or EHS to enroll

☒ Electrical Safety for Engineering Camino Course – contact EHS to enroll

☐ LiPo Battery Safety Training – contact MAKER Lab to enroll

☒ Review of SDS for chemicals involved in project – access SDS library at: chimeracloud.org/sds/

☒ Laboratory Specific Training – contact Lab or Shop Manager/Director

☒ Project Specific Training – contact Faculty Advisor

☒ Other (describe below):

Li-SOCl primary cell handling per the Saft datasheet. Open-water and boat safety.

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## EMERGENCY PROCEDURES

Life-threatening emergencies (police, fire, explosion, hazardous material spill/release, gas leak, etc.)

1. Call or text 911, then call SCU Campus Safety at 408-554-4444
2. Alert people in the vicinity and if necessary, activate the local alarm systems
3. Evacuate to your emergency assembly point (EAP). Fill in your EAP here: SCDI Courtyard
4. Remain nearby to advise emergency responders

For medical emergencies:

1. Stay with the individual if it is safe to do so. Do not move the individual unless absolutely necessary
2. Call SCU Campus Safety or 911
3. Administer first aid if you are trained to do so

If person received an electrical shock:

1. Do not touch the person
2. Shut off power and/or rescue the person with non-conductive object IF it is safe to approach
3. Call SCU Campus Safety or 911
4. Administer CPR or AED if trained to do so

If person exposed to hazardous materials:

1. Flush contamination from eyes/skin using the nearest emergency eyewash/shower for a least 15 minutes. Remove any contaminated clothing
2. Call SCU Campus Safety or 911

For building maintenance emergencies (power outages, plumbing leaks etc.): call Facilities 408-554-4742

*REPORT all injuries and incidents (including near misses) to faculty advisor and EHS*

## Project work outside of SCU campus

Generally, all project work must be performed at SCU campus. However, there may be few cases where work cannot be completed on campus (e.g. competition venue outside of SCU, no appropriate campus location to field-test). These exceptions must be described in the hazard assessment section above. Specific location(s) and their scope of work must have approval from the faculty advisor. Additional approvals may be required.

Check the appropriate project location scope:

◉ All project work will be performed on SCU campus

○ All project work cannot be completed on SCU campus. Provide more details below:

List offsite location(s) and provide description:

Kaneohe Bay, Hawaii (Mar to May 2027 deployment). Monterey/Santa Cruz. Kaneohe Yacht Club boat slip. All of these depend on permits and host approval, which are still in progress.

Describe supervision arrangements and emergency procedures for each offsite location:

Attended tests first. A buddy is always present. The host is HIMB or the pier or slip owner. All water work wears PFDs. Call 911 for an emergency. Notify the advisor and Environment, Health and Safety (EHS) after.

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## ACKNOWLEDGEMENT

By signing, I verify that:

1) I am aware of the hazards and risks of all the tasks associated with the project
2) I have received, or will receive all the necessary safety training, have read the safety manual and/or safety data sheets (SDSs) relevant to the project before performing any hazardous tasks.
3) I will follow all required safety precautions while working on this project, including but not limited to use of engineering controls, following safe work practices, and wearing appropriate personal protective equipment, as well as all emergency procedures

| Name of Project Team Member | Signature | Date |
|---|---|---|
| John Ryan Myrdal | John R Myrdal | 10/23/26 |
| Isabella Rodriguez | Isabella Rodriguez | 10/25/26 |
| David Chousal Cantu | | 10/24/26 |
| | | |
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## APPROVALS

This document must be reviewed and approved, in the order listed below, before any project work can begin. A copy of the approved document must be kept with the project team.

### 1. Faculty Advisor (required)

| Name: Jes Kuczenski<br>Department: GENG | Signature | Date |
|---|---|---|

### 2. Department Chair (required)

| Name | Signature | Date |
|---|---|---|
| Shoba Krishnan | | |

### 3. Laboratory Director/Manager (required if project involves use of lab or shop space)

| Name | Signature | Date |
|---|---|---|
| Navid Shaghaghi | Navid Shaghaghi | 10/23/2026 |

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### 4. EHS (required as determined by faculty advisor, department chair, or lab director/manager)

| Name | Signature | Date |
|---|---|---|
| | | |

### 5. Other (if requested by faculty advisor, department chair, lab director/manager, or EHS)

| Name | Signature | Date |
|---|---|---|
| | | |

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